Aga Khan University Examination Board

Notes from E-Marking Centre on HSSC-I Mathematics Examination April/May 2019

Introduction

This document has been produced for the teachers and candidates of Higher Secondary School Certificate (HSSC) part I Mathematics. It contains comments on candidates' responses to the 2019 HSSC-I Examination, indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on students' performance on every question and *some* specific examples of students' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that require candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Observations

In weaker responses, it was noted that the candidates failed to comprehend formulae and their applications according to given situations and did not score well in the examination. Candidates made mistakes in evaluating different trigonometric identities and also failed to read sine graph.
It was also noted that candidates failed to comprehend the concepts of Binomial theorem, Sequence and series but performed well in Matrices.

Detailed Comments:

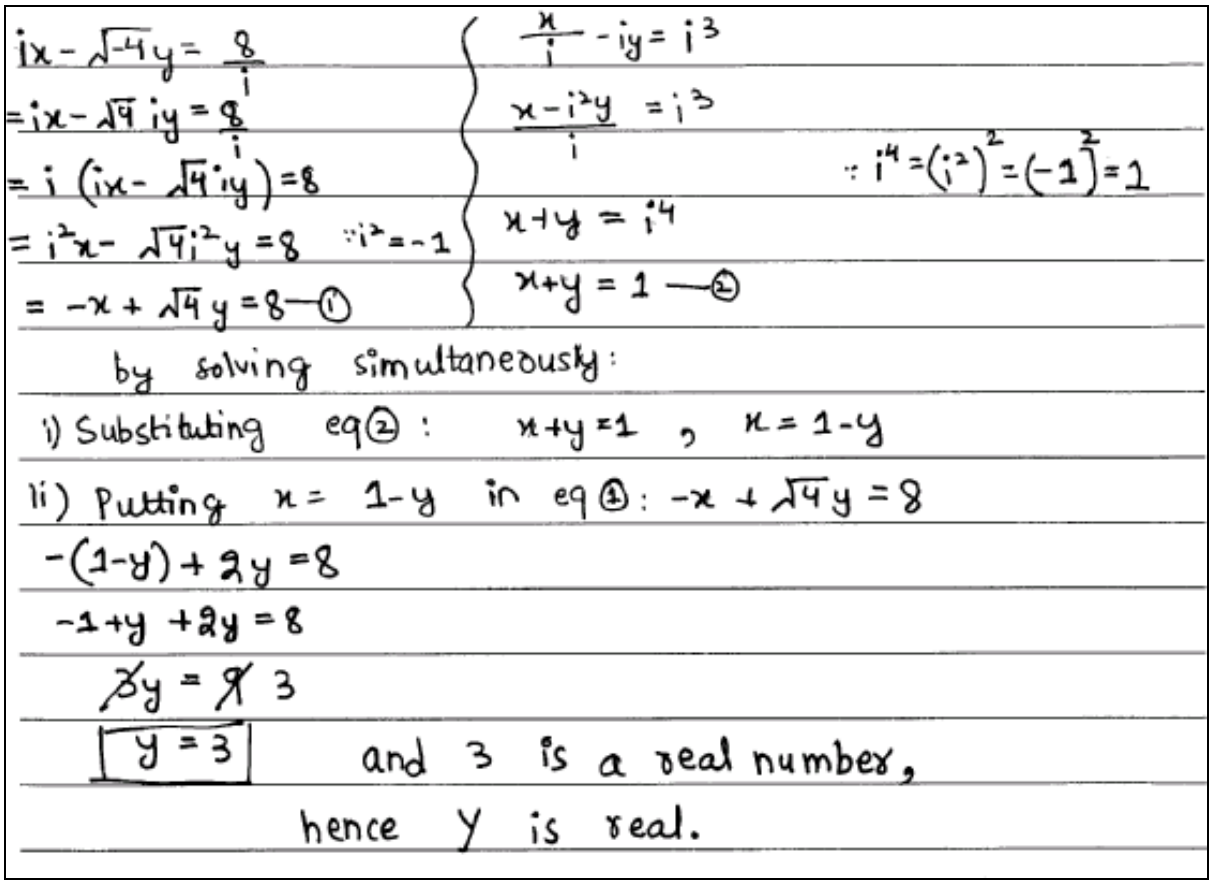
Constructed Response Questions (CRQs)

Question 1:

A pair of simultaneous linear equations with complex coefficients is given as $ix-\sqrt{-4}\,y=\frac{8}{i}$ and $\frac{x}{i}-iy=i^3$. Show that the value of y is real.

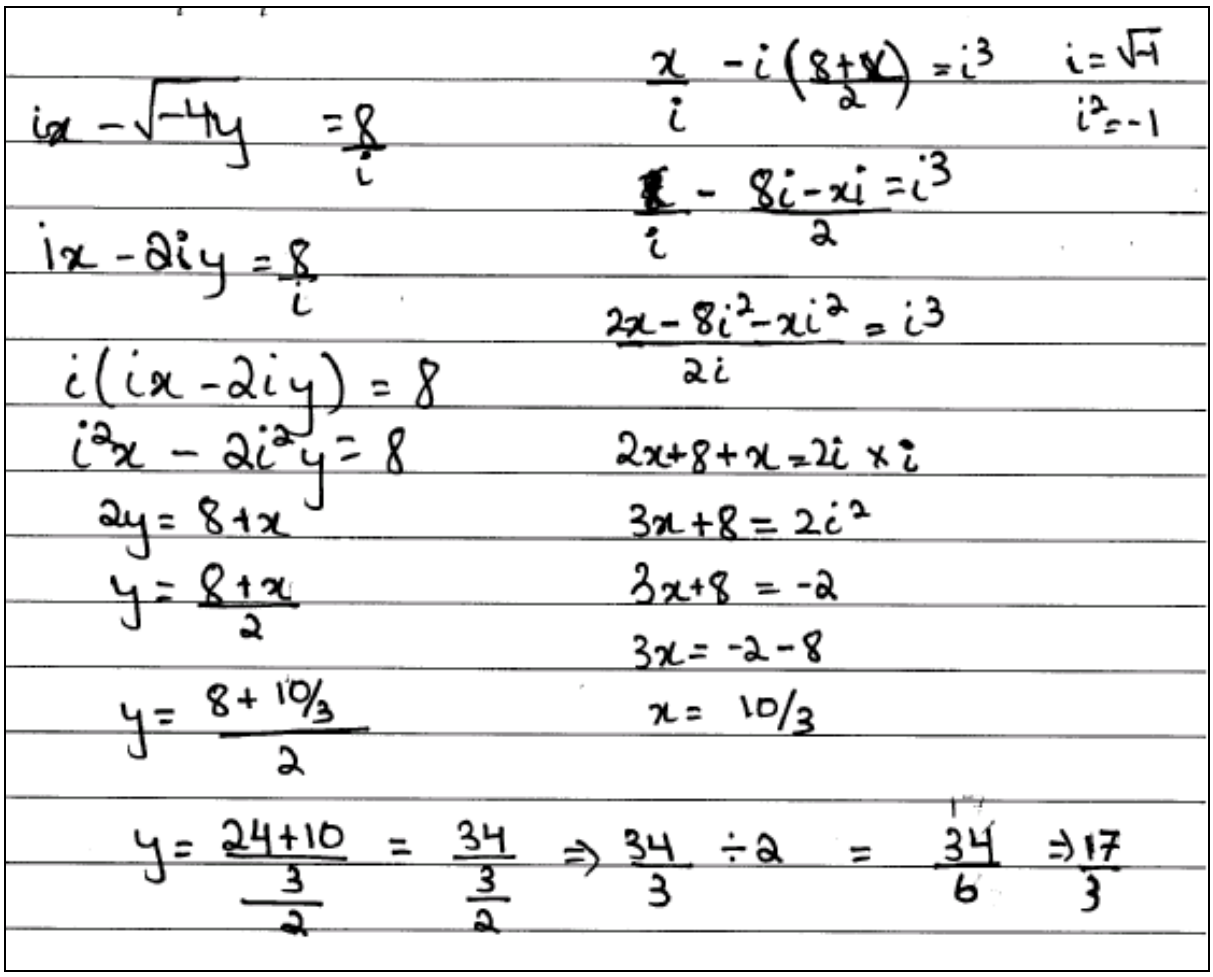
In *Better responses,* candidates simplified both the equations with complex coefficients very smartly as they were managed to write $x-2y=-8$ and $x+y=1$. This helped candidates to go further and therefore, they were able to show that the value of *y is* real.

Example:



Weaker responses reflected that the candidates had not clear understanding of complex numbers. That is why they made mistakes in solving the simultaneous equations with complex coefficients. It was seen that candidates wrote $\sqrt{-4}\,y$ as $-2y$ instead of $2yi$ that was actually incorrect. In addition, they failed to simplify the simultaneous equations with complex coefficients easily and as a result, they could not reach the required result.

Example:



Question 2a:

i. Under what condition(s) the given system of linear equations will become homogeneous?

$$k_{11}y_1 + k_{12}y_2 + k_{13}y_3 - m = 0$$
$$k_{21}y_1 + k_{22}y_2 + k_{23}y_3 - n = 0$$
$$k_{31}y_1 + k_{32}y_2 + k_{33}y_3 - p = 0$$

ii. It is given that $M = \begin{bmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{bmatrix}$ and I is an identity matrix of order 3.

Find the value of constant λ (lambda) when $M - \lambda I$ is a null matrix.

Better responses exhibited that candidates, in part **i**, well understood the question as this question had a very short answer and was purely a theoretical concept. Similarly, in part **ii**, candidates did well to use the condition and found the value of λ by simply subtracting the matrices and equating with the null matrix.

Example:

a.

i. Under what condition(s) the given system of linear equations will become homogeneous? (1 Mark)

$$k_{11}y_1 + k_{12}y_2 + k_{13}y_3 - m = 0$$
$$k_{21}y_1 + k_{22}y_2 + k_{23}y_3 - n = 0$$
$$k_{31}y_1 + k_{32}y_2 + k_{33}y_3 - p = 0$$

if $m = n = p = 0$ then the system of linear equation will become homogeneous.

ii. It is given that $M = \begin{bmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{bmatrix}$ and I is an identity matrix of order 3.

Find the value of constant λ (lambda) when $M - \lambda I$ is a null matrix. (2 Marks)

Given that

$$\begin{bmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} -3-\lambda & 0 & 0 \\ 0 & -3-\lambda & 0 \\ 0 & 0 & -3-\lambda \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore$ corresponding entries are equal

$\therefore \quad -3 - \lambda = 0$

$\boxed{\lambda = 3}$ Ans.

Weaker responses showed that candidates, in part **i**, failed to understand the question. In part **ii**, fewer candidates made mistakes in subtraction of matrices and it was also seen that candidates took λ as matrix rather than the constant λ. Although it was clearly mentioned in the question that λ is constant.

Example:

ii. It is given that $M = \begin{bmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{bmatrix}$ and I is an identity matrix of order 3.

Find the value of constant λ (lambda) when $M - \lambda I$ is a null matrix. (2 Marks)

$$\begin{bmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$-\lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$-\lambda = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\lambda = \begin{bmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{bmatrix}$$

Question 2b:

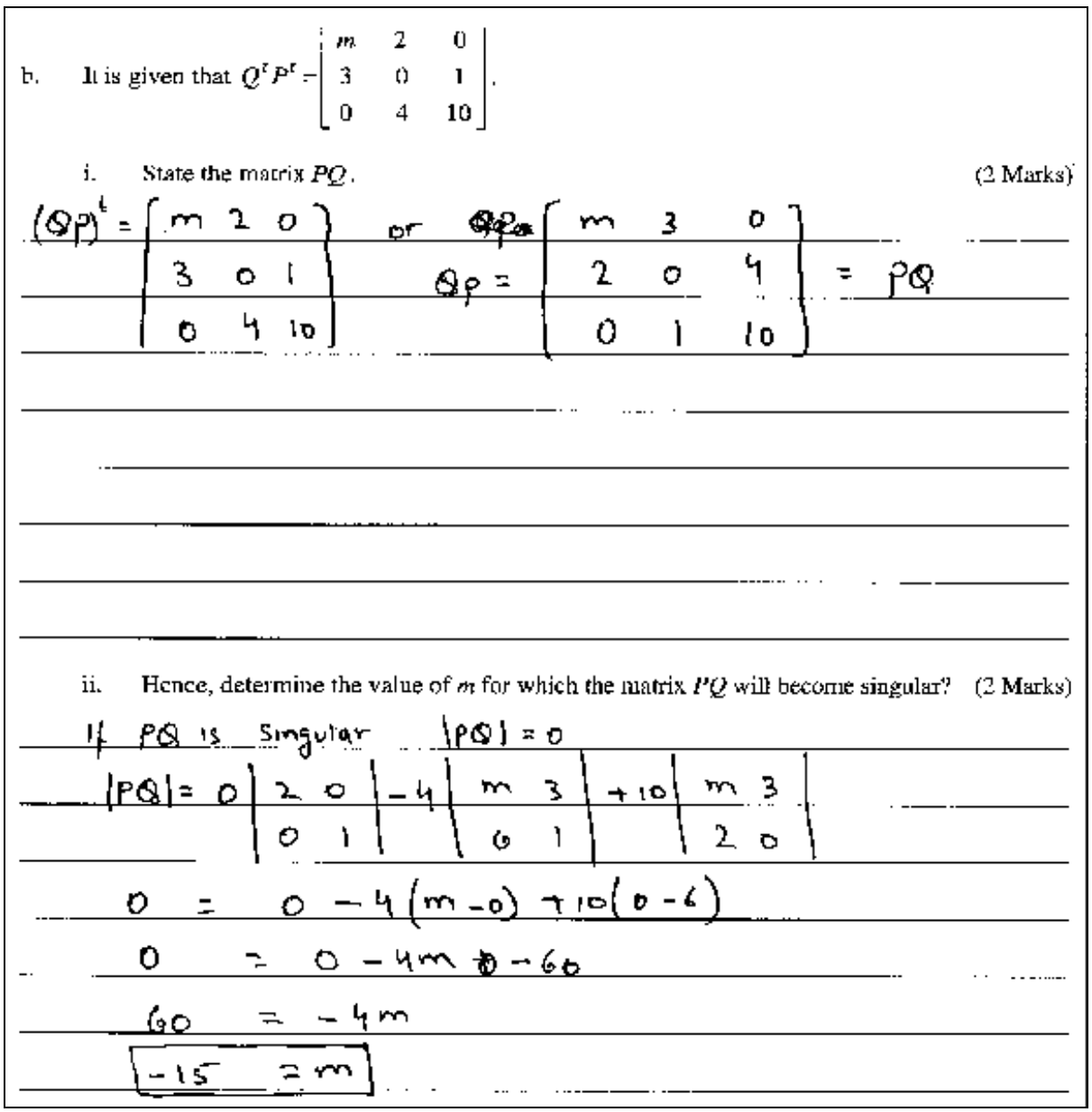
It is given that $Q^t P^t = \begin{bmatrix} m & 2 & 0 \\ 3 & 0 & 1 \\ 0 & 4 & 10 \end{bmatrix}$.

i. State the matrix PQ.

ii. Hence, determine the value of m for which the matrix PQ will become singular?

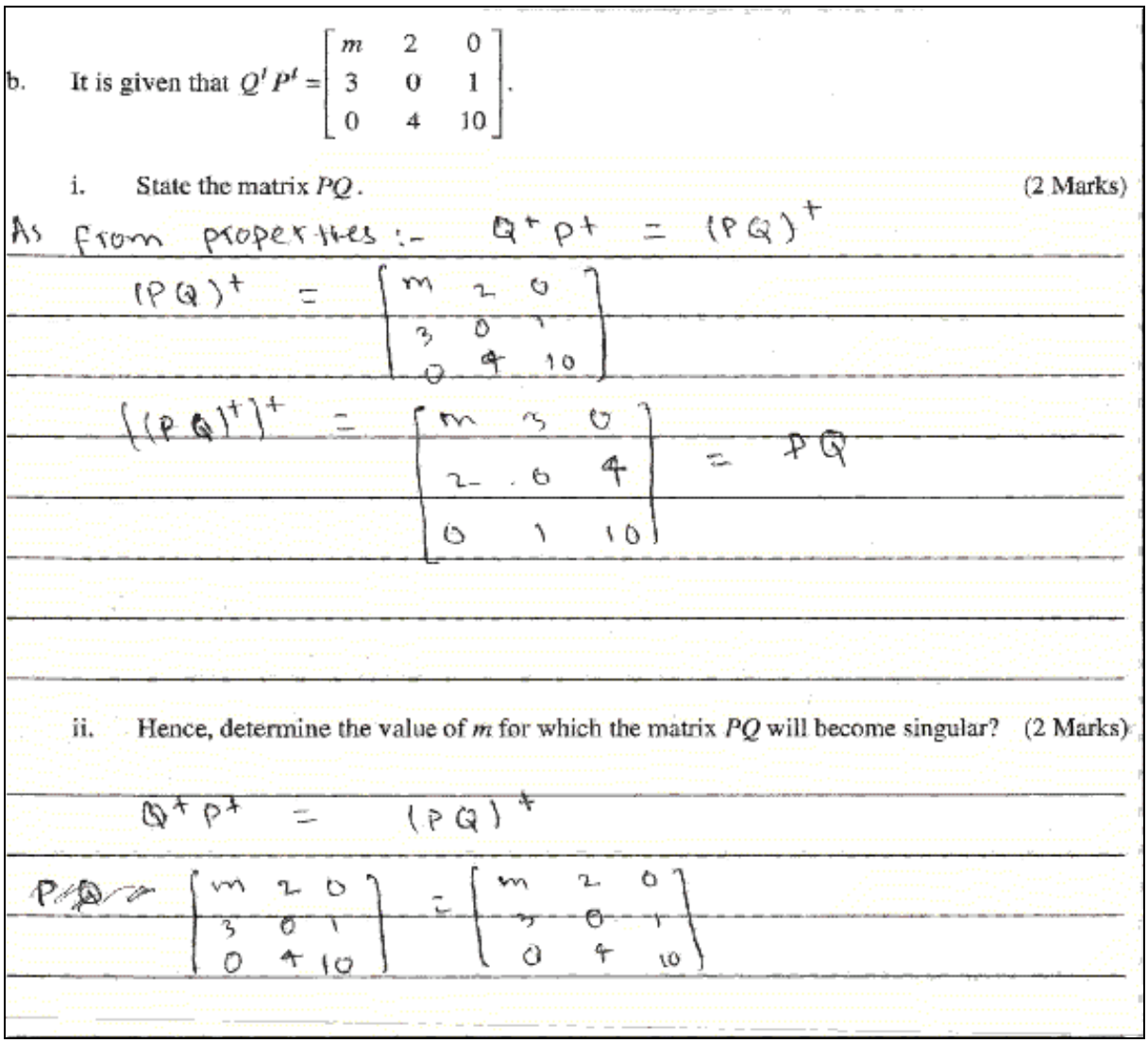
Better responses exhibited that candidates correctly applied the properties of transpose of matrix so that they were able to achieve the required matrix. As a result, in the next part, they were very much comfortable in finding the value of m by using the condition of singular matrix.

Example:



Weaker responses reflected lack of understanding of property $(AB)^t = B^tA^t$. Fewer candidates used correct property but made mistakes in using the condition of singular matrix which led to an incorrect value of *m*. It was also seen that candidates inappropriately applied the concept of determinant to find the required value.

Example:



b. It is given that $Q^t P^t = \begin{bmatrix} m & 2 & 0 \\ 3 & 0 & 1 \\ 0 & 4 & 10 \end{bmatrix}$.

i. State the matrix PQ. (2 Marks)

As from properties :- $Q^t P^t = (PQ)^t$

$(PQ)^t = \begin{bmatrix} m & 2 & 0 \\ 3 & 0 & 1 \\ 0 & 4 & 10 \end{bmatrix}$

$((PQ)^t)^t = \begin{bmatrix} m & 3 & 0 \\ 2 & 0 & 4 \\ 0 & 1 & 10 \end{bmatrix} = PQ$

ii. Hence, determine the value of m for which the matrix PQ will become singular? (2 Marks)

$Q^t P^t = (PQ)^t$

$PQ = \begin{bmatrix} m & 2 & 0 \\ 3 & 0 & 1 \\ 0 & 4 & 10 \end{bmatrix} = \begin{bmatrix} m & 2 & 0 \\ 3 & 0 & 1 \\ 0 & 4 & 10 \end{bmatrix}$

Question 3a:

The arithmetic progression (A.P) and geometric progression (G.P) are defined in such a way that the first term, fourth term and ninth term of an A.P will form a G.P.

i. Express, in terms of a and d, the first, fourth and ninth terms of A.P.
ii. Use the given condition to show that $2a = 9d$.
iii. Hence, find the value of tenth term of an A.P. when $a = 4$.

Better responses exhibited that candidates applied the conditions quite well to write the required terms of A.P. and consequently converted into G.P. Because of this, they were able to reach $2a = 9d$. Finally, they got the tenth term as well either by using the above relation or just by finding the value of d.

Example:

a. The arithmetic progression (A.P) and geometric progression (G.P) are defined in such a way that the first term, fourth term and ninth term of an A.P will form a G.P.

i. Express, in terms of a and d, the first, fourth and ninth terms of A.P. (1 Mark)

$a_1 = a$ | $a_9 = a + 8d$

$a_4 = a + 3d$

ii. Use the given condition to show that $2a = 9d$. (3 Marks)

If $a_1, a_4, a_9 \ldots$ are in G.P

then $\frac{a_4}{a_1} = \frac{a_9}{a_4}$

$(a_4)^2 = a_1 \times a_9$

$(a+3d)^2 = a(a+8d)$

$a^2 + 6ad + 9d^2 = a^2 + 8ad$

$a^2 - a^2 + 9d^2 = 8ad - 6ad$

$9d^2 = 2ad$

$\boxed{9d = 2a}$ Proved.

iii. Hence, find the value of tenth term of an A.P. when $a = 4$. (1 Mark)

$9d = 2a$ | $d = \frac{8}{9}$ | $a_{10} = a_1 + 9d$ | $a_{10} = 4 + 8$

$9d = 2(4)$ | | $a_{10} = 4 + 9(\frac{8}{9})$ | $\boxed{a_{10} = 12}$ Ans.

Weaker responses reflected that candidates even though wrote the required terms of A.P. but failed in translating given conditions in the form of G.P. and hence, they were unable to reach the required proof and the tenth term. It was also seen that some candidates used A.P. only without converting in G.P. As a result, they could not find the desired result.

Example:

a. The arithmetic progression (A.P) and geometric progression (G.P) are defined in such a way that the first term, fourth term and ninth term of an A.P will form a G.P.

i. Express, in terms of a and d, the first, fourth and ninth terms of A.P. (1 Mark)

$T_1 = a_1$, $T_4 = a_1 + 3d$ $T_9 = a_1 + 8d$

$a_4 = a_1 + 3d$, $a_9 = a_1 + 8d$

ii. Use the given condition to show that $2a = 9d$. (3 Marks)

to prove, $2a = 9d$

$a = \frac{9d}{2}$ → i.

$a_4 = a_1 + 3d$

$a_4 = \frac{9d}{2} + 3d$

$a_4 = \frac{9d + 6d}{2}$

$a_4 = \frac{15d}{2}$

$a_9 = a_1 + 8d$ $\therefore a = \frac{9d}{2}$

for $a_9 = \left(\frac{9d}{2}\right) + 8d$

$= \frac{9d + 16d}{2}$

$= \frac{25d}{2}$

now,

$a_9 - a_4 = a_1 + 8d - (a_1 + 3d)$

$= \frac{25d}{2} - \frac{15d}{2} = a_1 - a_1 + 5d$

$= \frac{10d}{2} = 5d$

for $a_4 = \left(\frac{9d}{2}\right) + 3d = 5d = 5d$.

iii. Hence, find the value of tenth term of an A.P. when $a = 4$. (1 Mark)

$a_{10} = a_1 + 9d$

as $2a = 9d$

$2(4) = 9d$

$d = \frac{8}{9}$

$\therefore a_{10} = a_1 + 9d$

$= 4 + 9\left(\frac{8}{9}\right) = 12$

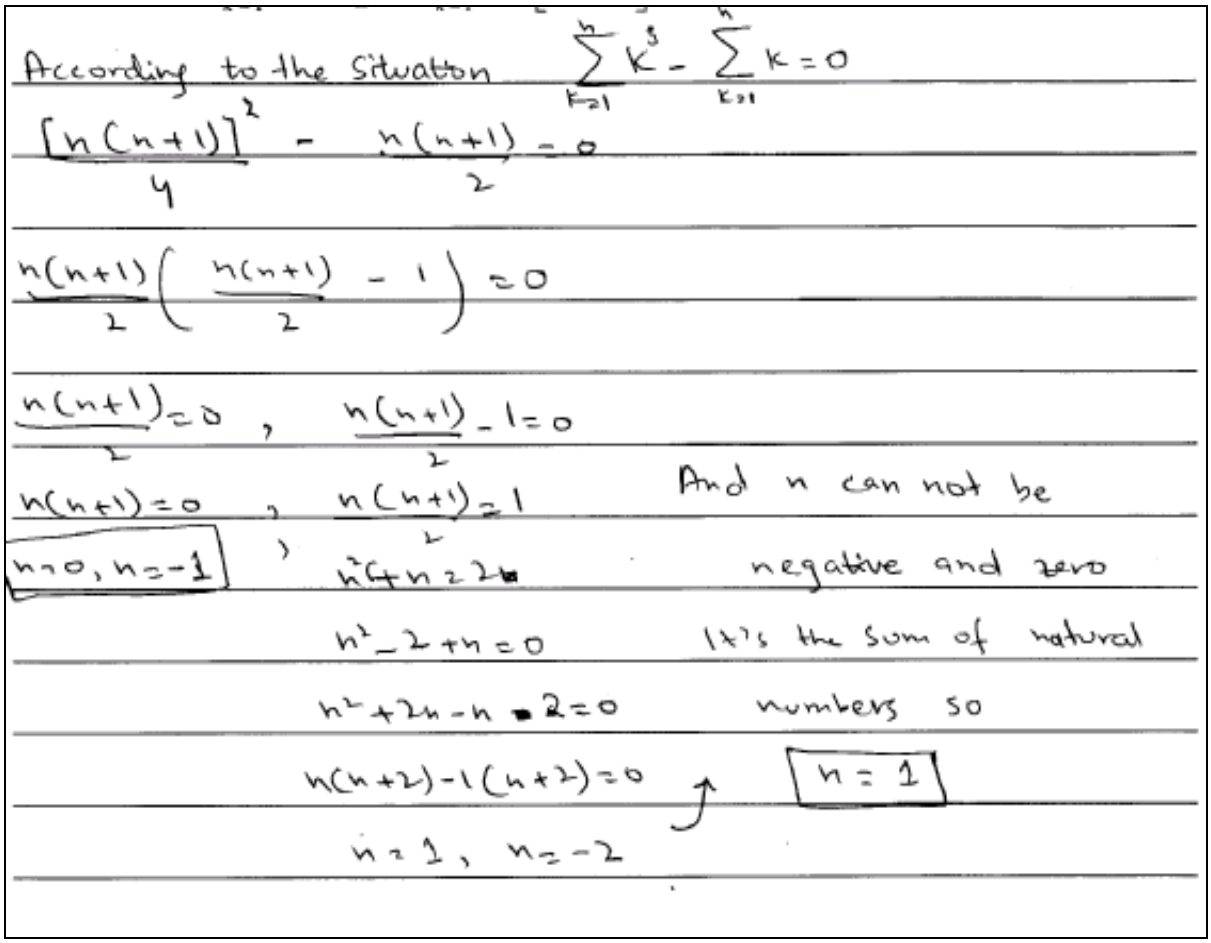
Question 3b:

The sum of first *n* natural numbers $\sum_{k=1}^{n} k$ and the sum of the cubes of first *n* natural numbers $\sum_{k=1}^{n} k^3$ differ by zero. Find the value of *n*.

(**Formulae**: $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$, $\sum_{k=1}^{n} k^3 = \left[\frac{n(n+1)}{2}\right]^2$)

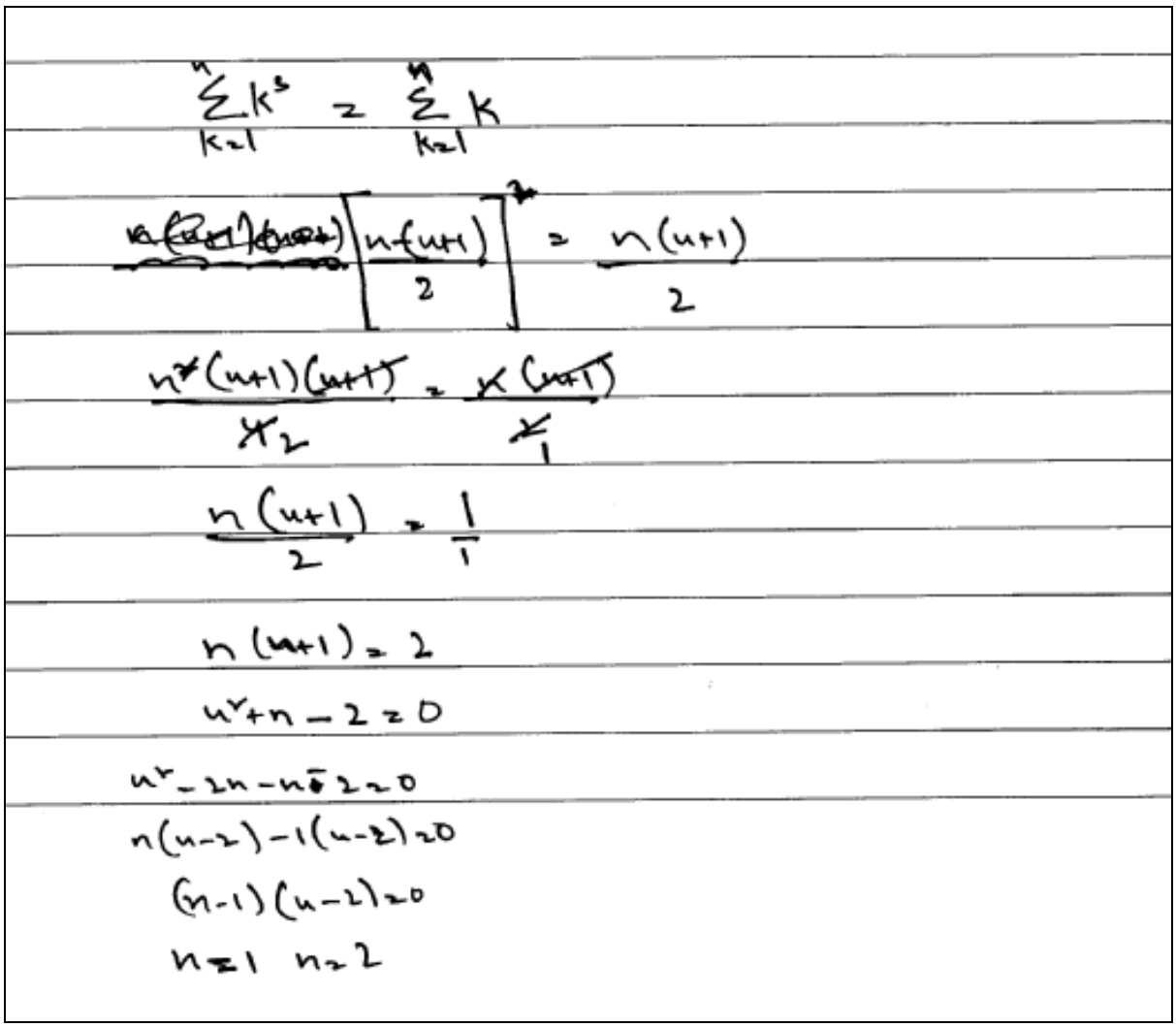
Better responses revealed that candidates well understood the question by writing $\frac{n(n+1)}{2} - \left[\frac{n(n+1)}{2}\right]^2 = 0$. This helped candidates to move further in steps and simplified the equations correctly to achieve the value of *n*.

Example:



Weaker responses demonstrated that candidates initially failed to translate the given condition into mathematical statement. Subsequently, they were unable to get the correct answer. It was also evident that candidates made mistakes in simplification and as a result they got the negative value of n which was definitely unacceptable because n is a natural number.

Example:

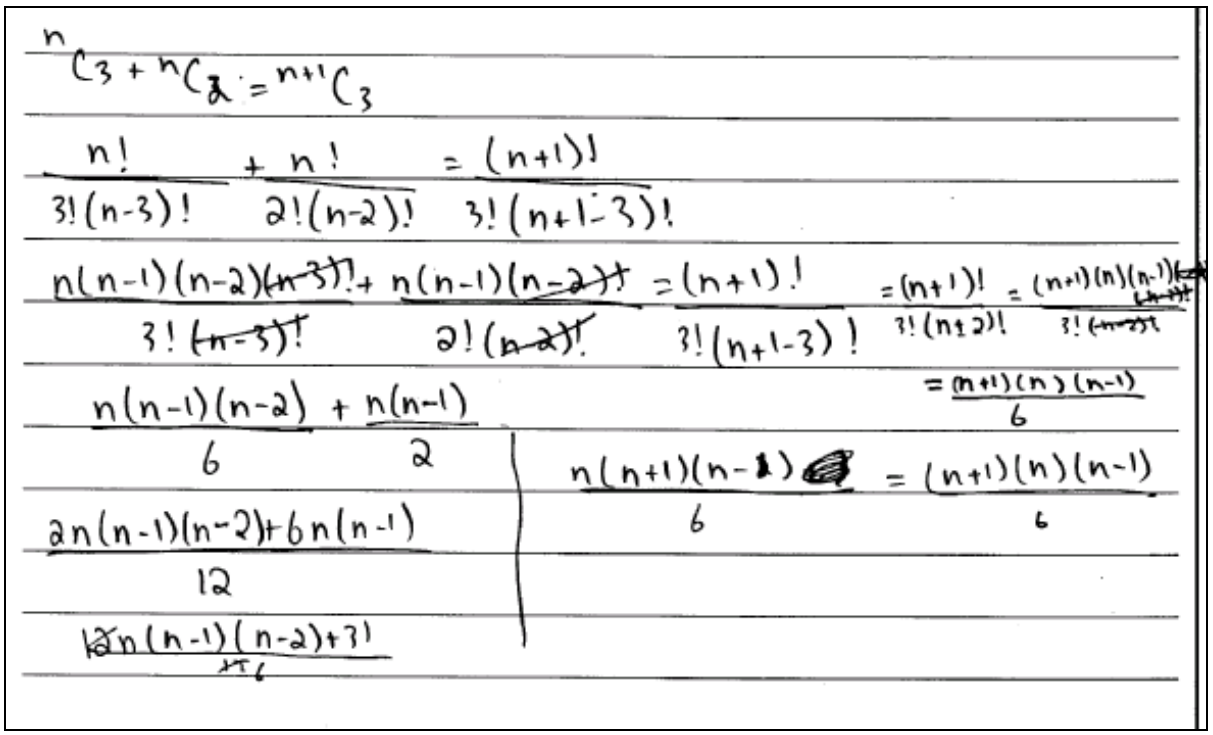


Question 4:

Prove that $\binom{n}{3}+\binom{n}{2}=\binom{n+1}{3}$, where *n* is a positive integer.

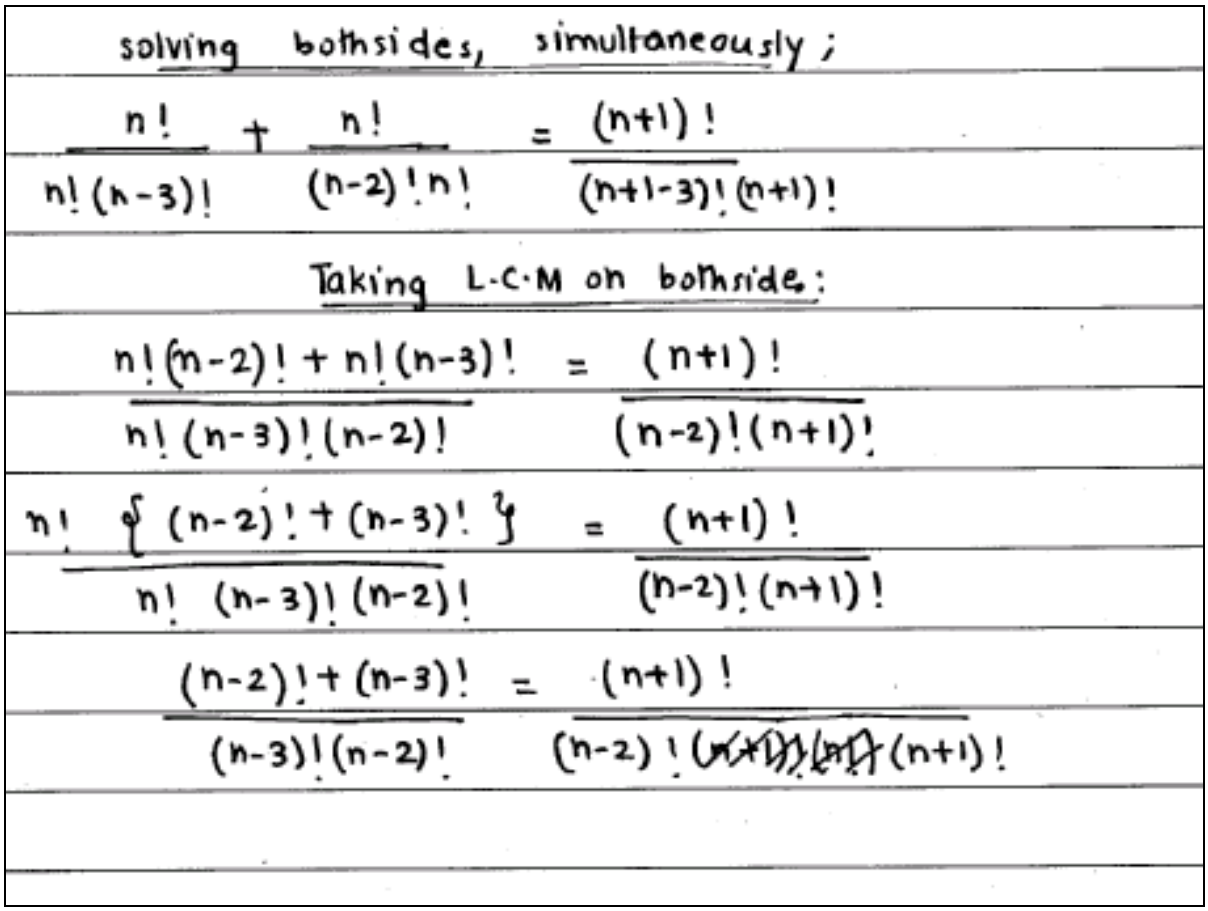
Better responses indicated that the candidates well understood the question and correctly used the formula of $\binom{n}{r}$ to evaluate $\binom{n}{3}$, $\binom{n}{2}$ and $\binom{n+1}{3}$. This helped candidates to satisfy both the sides by correct simplification.

Example:



Weaker responses reflected that candidates even though used the formula correctly but got stuck in further steps because mistakes occurred while simplifying $\binom{n}{3}$, $\binom{n}{2}$ and $\binom{n+1}{3}$. Because of this, they were unable to satisfy both the sides and could not get the desired result.

Example:



Question 5:

(**Formula**: $T_{r+1} = {}^nC_r a^{n-r} b^r$, where, symbols have their usual meanings.)

In the binomial expansion of $\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^n$,

a. show that $r = \frac{n}{2}$ for which the term is constant.

b. hence, find the value of *n* and of *k* if the constant term is given by ${}^kC_3$.

Better responses exhibited that candidates correctly substituted the values in the formula and simplified accordingly to get $r = \frac{n}{2}$. It was also seen that few candidates directly substituted $r = \frac{n}{2}$ in the binomial formula and finally got the constant term. Hence, they were able to proceed further in order to find the value of *n* and of *k* by comparing ${}^nC_r = {}^kC_3$.

Example:

a. show that $r = \frac{n}{2}$ for which the term is constant. (3 Marks)

General term

$T_{r+1} = {}^nC_r a^{n-r} b^r$

$T_{r+1} = \binom{n}{r}(x^{1/2})^{n-r}\left(\frac{1}{x^{1/2}}\right)^r$

$T_{r+1} = \binom{n}{r}(x^{1/2})^{n-r}(x^{1/2})^r$

$T_{r+1} = {}^nC_r x^{1/2n - 1/2r - 1/2r} \cdot (1)^r$

$T_{r+1} = {}^nC_r x^{1/2n - r}$

If x has power = 0 the term will be a constant so

$1/2n - r = 0 \quad 1/2n = r$

$r = n/2$ → proved.

b. hence, find the value of n and of k if the constant term is given by ${}^kC_3$. (4 Marks)

If the constant term is given by ${}^kC_3$ it means that $r = 3$. Putting the value of r in the above relation $r = n/2$

$n = (3)(2)$

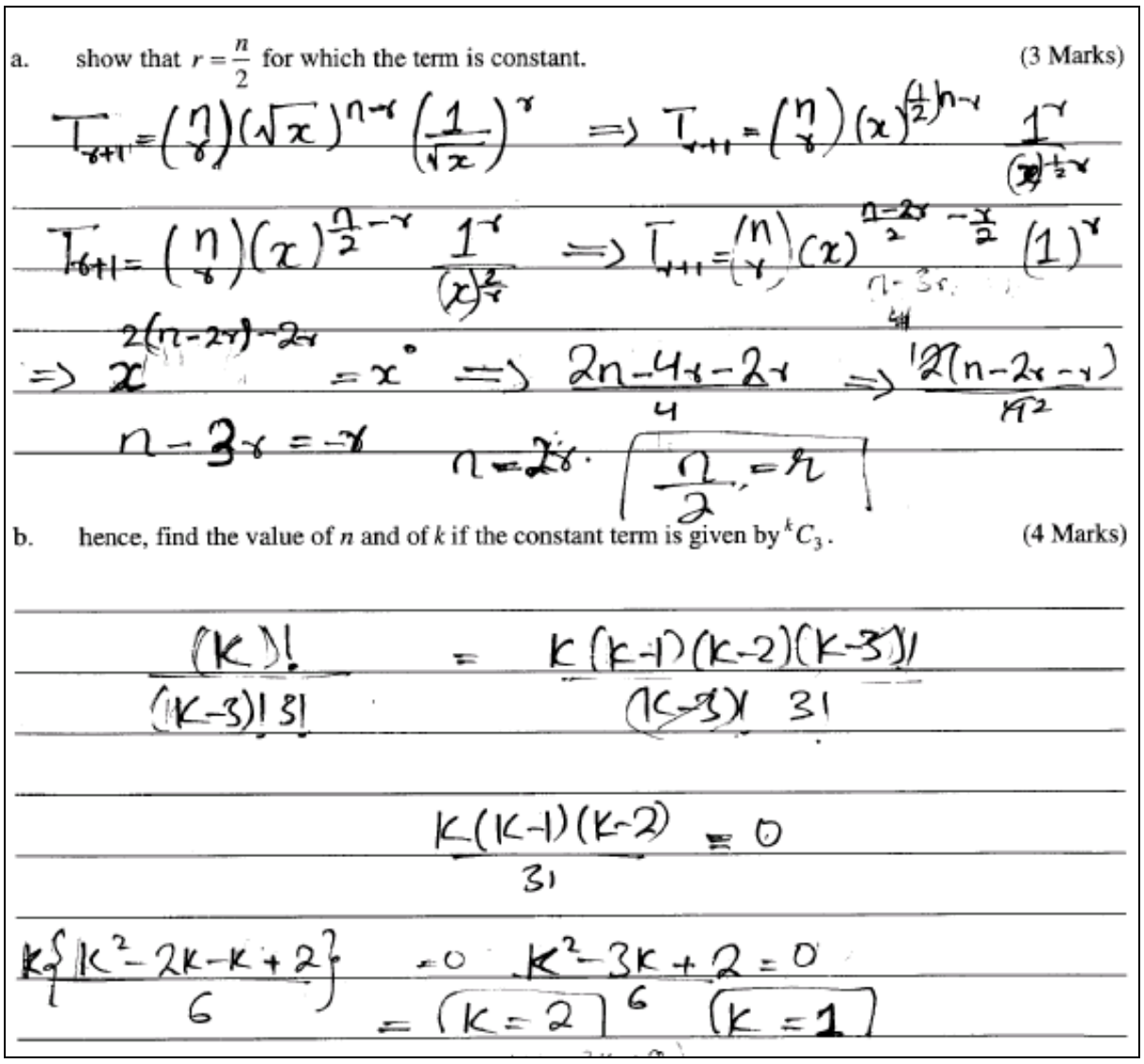
$n = 6$

Such that the value of $k = n$

So $k = 6$ so the constant term will be equal to (1)

Weaker responses indicated that candidates faced difficulty to understand the question and they were unable to substitute the values in the given formula. As a result, they could not reach the desired result. It was also seen that after substituting the values in the formula they were failed in simplification further to get the required proof. Some candidates just evaluated ${}^{k}C_{3}$ by using the formula of ${}^{n}C_{r}$ but after that they got confused in further calculation.

Example:



Question 6:

This question offered a choice between part **a** and part **b**. Candidates chose to attempt part **a** more than part **b**.

Question 6a:

The length x of one side of a square is decreased by 8 cm and another by 4 cm to form a rectangle.

i. Find an expression, in terms of x, for

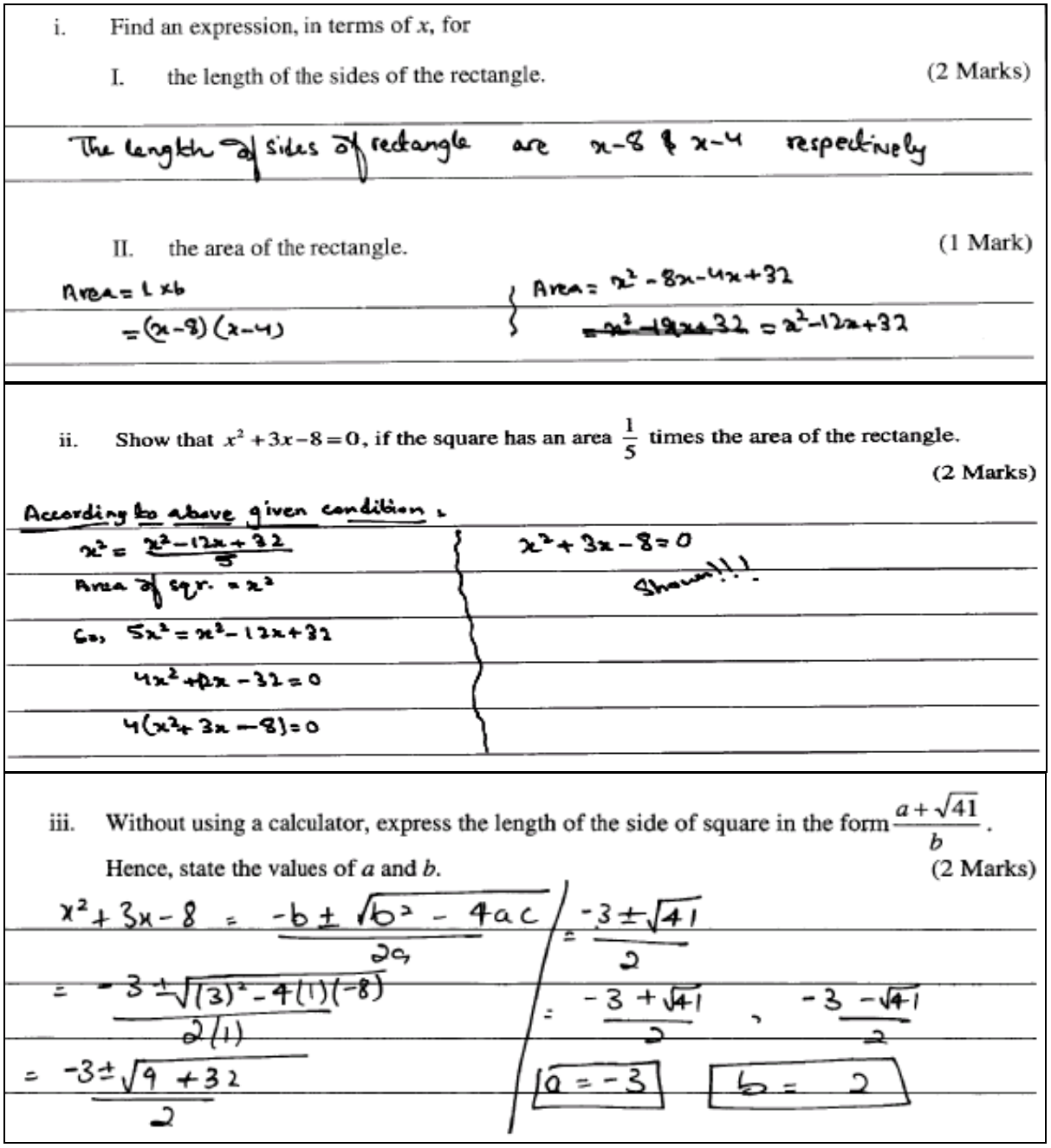
   I. the length of the sides of the rectangle.
   II. the area of the rectangle.

ii. Show that $x^2 + 3x - 8 = 0$, if the square has an area $\frac{1}{5}$ times the area of the rectangle.

iii. Without using a calculator, express the length of the side of square in the form $\frac{a + \sqrt{41}}{b}$. Hence, state the values of a and b.

Better responses reflected that candidates transformed the word problems into mathematical statement very well. This helped them, in part **i**, to get expressions for the length of sides and area of rectangle. In part **ii**, they were able to manage the required equation with the help of the given condition. Similarly, in part **iii**, candidates found an expression for the length of the side of square by applying the quadratic formula and hence, stated the values of a and b easily.

Example:



i. Find an expression, in terms of x, for

I. the length of the sides of the rectangle. (2 Marks)

The length of sides of rectangle are $x-8$ & $x-4$ respectively

II. the area of the rectangle. (1 Mark)

Area = l × b

$= (x-8)(x-4)$

Area = $x^2 - 8x - 4x + 32$

$= x^2 - 12x + 32$

ii. Show that $x^2 + 3x - 8 = 0$, if the square has an area $\frac{1}{5}$ times the area of the rectangle. (2 Marks)

According to above given condition,

$x^2 = \frac{x^2 - 12x + 32}{5}$

Area of sqr. $= x^2$

So, $5x^2 = x^2 - 12x + 32$

$4x^2 + 12x - 32 = 0$

$4(x^2 + 3x - 8) = 0$

$x^2 + 3x - 8 = 0$

Shown!!!

iii. Without using a calculator, express the length of the side of square in the form $\frac{a+\sqrt{41}}{b}$.

Hence, state the values of a and b. (2 Marks)

$x^2 + 3x - 8 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$= \frac{-3 \pm \sqrt{(3)^2 - 4(1)(-8)}}{2(1)}$

$= \frac{-3 \pm \sqrt{9 + 32}}{2}$

$= \frac{-3 \pm \sqrt{41}}{2}$

$= \frac{-3 + \sqrt{41}}{2}, \frac{-3 - \sqrt{41}}{2}$

$a = -3$ $b = 2$

Weaker responses indicated that candidates were clueless to convert the given word problem into mathematical statement to solve it further. As a result, they could not get the required equation successfully which resulted in the loss of marks. It was seen that fewer candidates just managed to express the length of the side of square in the required form by applying quadratic formula or some of them solved completely with the help of calculator, although calculator for this particular part is not allowed.

Example:

i. Find an expression, in terms of x, for

I. the length of the sides of the rectangle. (2 Marks)

Solution => $8-x = 4-x$ $\Rightarrow 8-4 = x-x$ $12 = 2x$ $x = 6$

$8-x = x-4 \Rightarrow 8+4 = x+x \Rightarrow 12/2 = x$

II. the area of the rectangle. (1 Mark)

Solution => $b^2 \times b^2$

$\Rightarrow$ Area $= 6 \times 6 = 36\,cm^2$

ii. Show that $x^2+3x-8=0$, if the square has an area $\frac{1}{5}$ times the area of the rectangle. (2 Marks)

area of square $\Rightarrow \frac{1}{5}(x^2-12x+13)=0$

$\Rightarrow \frac{x^2}{5} - \frac{12x}{5} + \frac{13}{5} = 0$

$x^2+3x-8=0$

$x = \frac{-3 \pm \sqrt{41}}{2}$.

iii. Without using a calculator, express the length of the side of square in the form $\frac{a+\sqrt{41}}{b}$.

Hence, state the values of a and b. (2 Marks)

$a = 1, \; b = 3$.

$\frac{1+\sqrt{41}}{3}$

$= \frac{1+\sqrt{41}}{3}$.

Better responses of part **i** demonstrated that candidates substituted the value of *p, q* and *r* very well in $p-2q+r^2$. Consequently, they applied the properties of cube roots of unity $\omega^3=1$ and $\omega^2+\omega+1=0$ followed by correct simplification and got the required answer. Similarly, in part **ii**, candidates displayed good skills to simplify the given expression. As a result, they got the value of required expression an imaginary number by applying again the properties of cube roots of unity.

Question 6b:

Given that $p=\frac{4}{\omega}, q=-\frac{2}{\omega^2}$ and $r=\frac{1}{\omega^9}$, show that

(Note: ω and ω^2 are the complex cube roots of unity.)

i. $p-2q+r^2$ is a real number.

ii. $\sqrt{p^2+(2q)^2}$ is an imaginary number.

Example:

i. $p-2q+r^2$ is a real number. (3 Marks)

$= \frac{4}{w} - 2\left(-\frac{2}{w^2}\right) + \left(\frac{1}{w^9}\right)^2$ $\quad \frac{4w+4+w^2}{w^2}$

$= \frac{4}{w} + \frac{4}{w^2} + \frac{1}{w^{18}}$ $\quad \frac{4w+4-1-w}{w^2}$ $\quad \therefore 1+w+w^2=0$, $\therefore w^2=-1-w$

$= \frac{4}{w} + \frac{4}{w^2} + \frac{1}{(w^3)^6}$ $\quad \frac{3w+3}{w^2}$ $\quad 1+w+w^2=0$

$\because w^3=1$

$= \frac{4}{w} + \frac{4}{w^2} + \frac{1}{1}$ $\quad \frac{3(w+1)}{w^2}$ $\quad 1+w=-w^2$

$= \frac{4w+4+w^2}{w^2}$ $\quad \frac{3(-w^2)}{w^2} = -3$ Ans

ii. $\sqrt{p^2+(2q)^2}$ is an imaginary number. (4 Marks)

$\sqrt{\left(\frac{4}{w}\right)^2+\left[2\left(\frac{-2}{w^2}\right)\right]^2}$ $\quad \sqrt{\frac{-16w}{w^4}}$

$\sqrt{\frac{16}{w^2}+\frac{16}{w^4}}$ $\quad \sqrt{\frac{-16}{w^3}}$ as $w^3=1$

$\sqrt{\frac{16w^3+16}{w^4}}$ $\quad \sqrt{-16}$

$\sqrt{\frac{16(1+w^2)}{w^4}}$ $\quad \Rightarrow \pm 4i$ (proved)

It is an imaginary number.

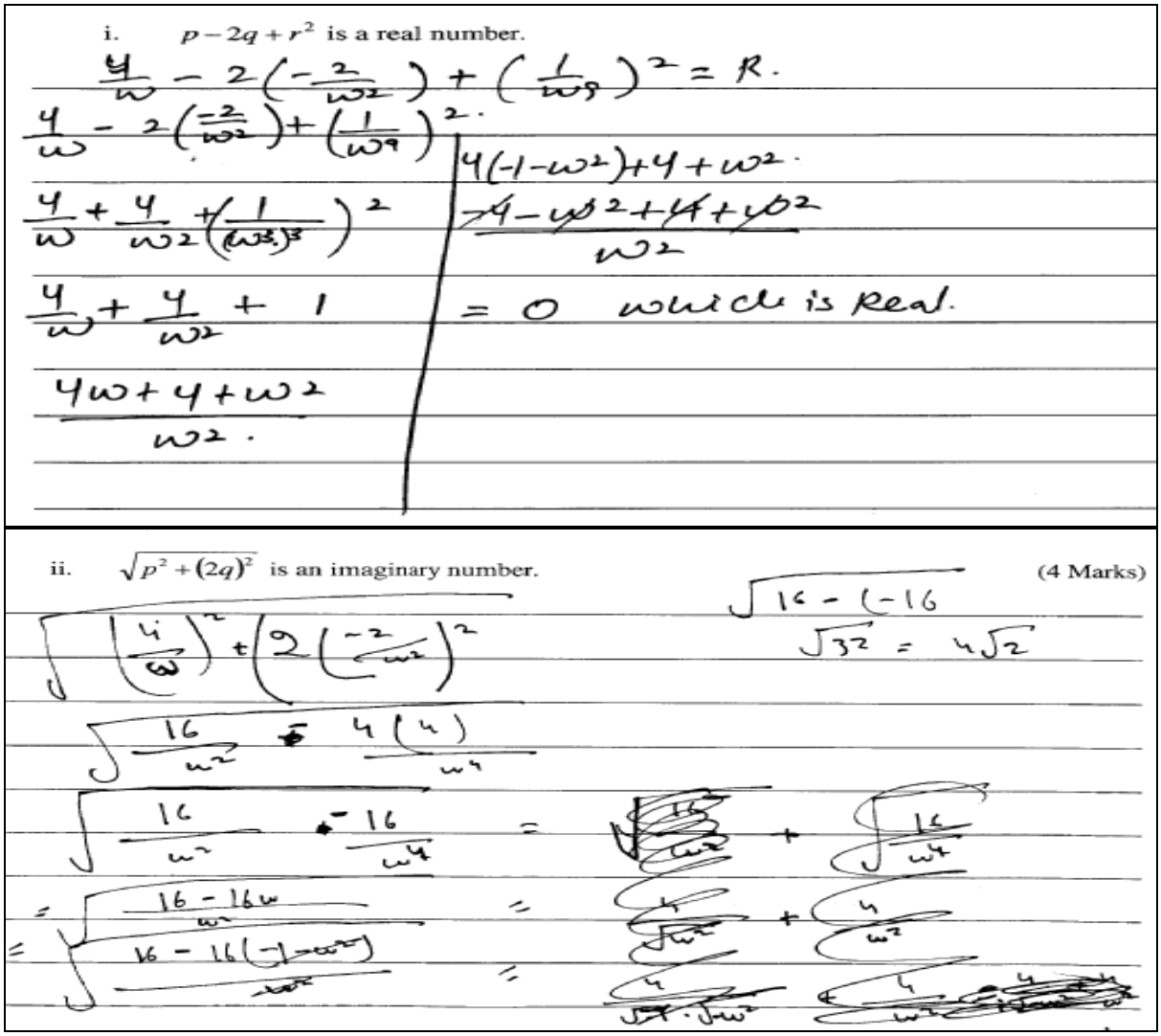
Weaker responses showed that in part **i**, the candidates even though substituted the values correctly but failed apply the properties of cube roots of unity accordingly. Consequently, they were unable to get the desired answer.

In part **ii**, weaker responses revealed that candidates failed to comprehend or apply the properties of cube roots of unity. They also showed mistakes in the process of simplification to get the required results. Few examples of mistakes which were done by the candidates are given below:

- $\sqrt{p^2+(2q)^2}$ as $p+2q$
- $\sqrt{p^2+(2q)^2}$ as $\sqrt{p+2q^2}$
- $\sqrt{p^2+(2q)^2}$ as $\sqrt{p+4q}$

Because of these mistakes, candidates were unable to go further in order to get the required result which resulted in the loss of marks.

Example:



Question 7:

This question offered a choice between part **a** , part **b** and part **c**. Candidates chose to attempt part **a** and part **c** more than part **b**.

Question 7a:

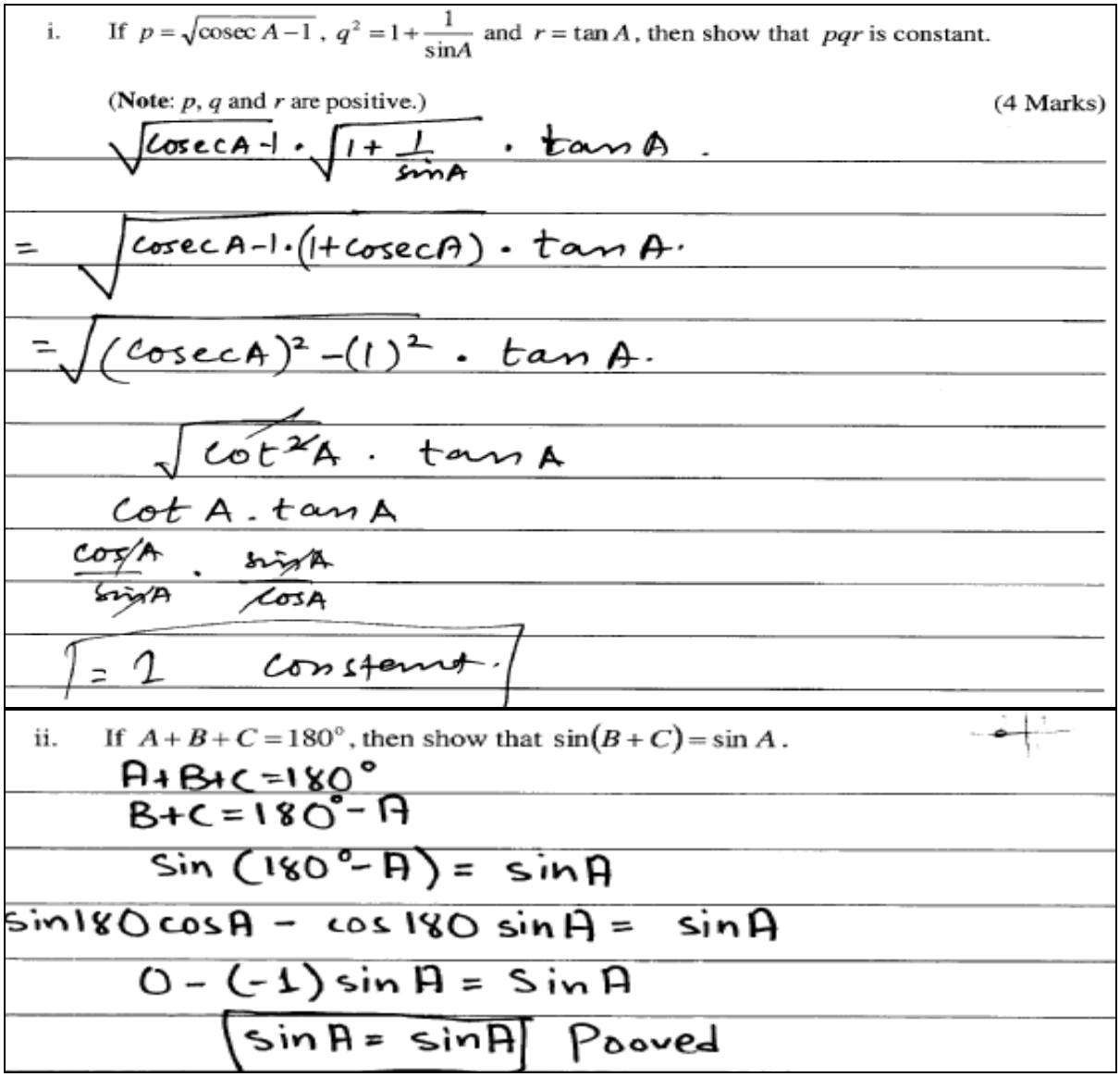
i. If $p = \sqrt{\text{cosec}\, A - 1}$, $q^2 = 1 + \dfrac{1}{\sin A}$ and $r = \tan A$, then show that pqr is constant.
(**Note**: p, q and r are positive.)

ii. If $A + B + C = 180^\circ$, then show that $\sin(B + C) = \sin A$.

Better responses indicated that candidates, in part **i**, substituted the values of p, q and r by finding $q = \sqrt{1 + \text{cosec}\, A}$ correctly as well. As a result, candidates applied the correct trigonometric identities aptly such as $\cot\theta = \pm\sqrt{\text{cos}\, ec^2\theta - 1}$ or $\text{cosec}\,\theta = \dfrac{1}{\sin\theta}$ to get the required result.

Similarly, in part **ii**, candidates used the given condition $A + B + C = 180^\circ$ very well by writing $B + C = 180^\circ - A$ and applied sine on both sides to achieve the required form. It was also seen that fewer candidates replaced $B + C$ by $180^\circ - A$ and applied the formula of $\sin(\alpha + \beta)$ correctly to get the other side i.e. sin A.

Example:



Weaker responses reflected that the candidates, in part **i**, applied wrong trigonometric identities or formulae. They failed in substituting the values of *p*, *q* and *r* especially simplified incorrect value of *q*. Because of this, they were unable to use the right approach to go further in calculation and hence, they could not reach the required result.

In part **ii**, candidates were clueless in using the given relation $A+B+C=180^o$ and therefore, they were unable to get the desired form

Example:

i. If $p=\sqrt{\operatorname{cosec} A-1}$, $q^2=1+\frac{1}{\sin A}$ and $r=\tan A$, then show that pqr is constant.

(**Note**: p, q and r are positive.) (4 Marks)

$q^2 = 1 + \operatorname{cosec} A$

$q = \sqrt{1+\operatorname{cosec} A}$

$q = \sqrt{\operatorname{cosec} A + 1}$

$p = \sqrt{\operatorname{cosec} A - 1}$

$r = \tan A$

$p \times q \times r$

$\sqrt{\operatorname{cosec} A - 1} \times \sqrt{\operatorname{cosec} A + 1} \times \tan A$

$\operatorname{cosec}^2 A - 1 \times \tan^2 A$

$\left(\frac{1}{\sin^2 A} - 1\right) \times \frac{\sin^2 A}{\cos A}$

$\frac{\sin A}{\sin^2 A \cos A} - \frac{\sin A}{\cos A}$

$\frac{1}{\cos A} \cdot \left[\operatorname{cosec}^2 - \sin A\right]$

$\frac{1}{\cos A}\left[\frac{1}{\sin A} - \sin A\right]$ $\Leftarrow$ $\frac{1}{\cos A}\left[\frac{1-\sin^2 A}{\sin A}\right]$

$\frac{1}{\cos A}\left[\frac{\cos^2 A}{\sin A}\right] = \cot A$

ii. If $A+B+C=180^\circ$, then show that $\sin(B+C)=\sin A$. (3 Marks)

$\sin(B+C) = \sin B \cos C + \cos B \sin C$

$\sin 60 \cos 60 + \cos 60 \sin 60 = \sin 60$

$\frac{\sqrt{3}}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}$

Question 7b:

Without using a calculator, evaluate the following:

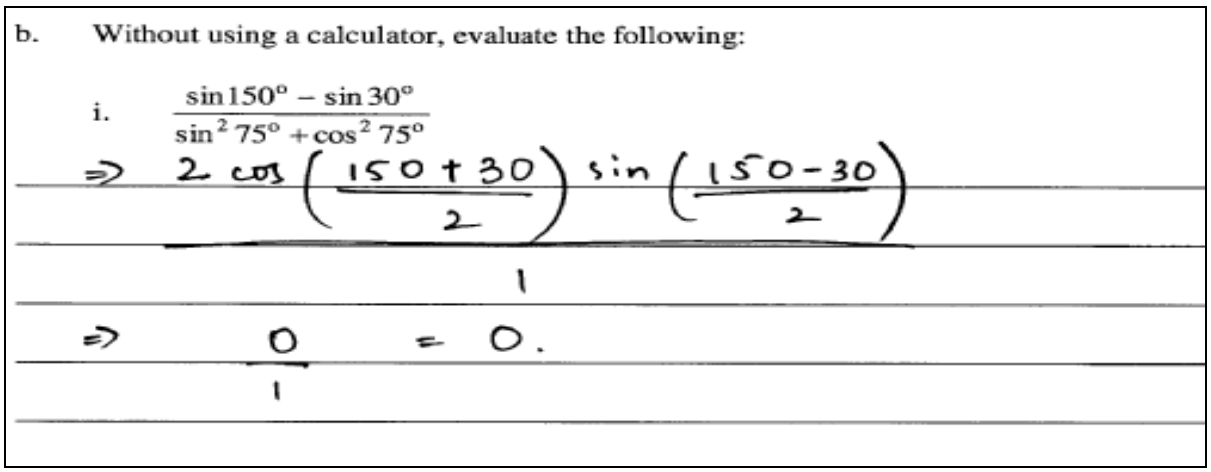
i. $\dfrac{\sin 150^\circ - \sin 30^\circ}{\sin^2 75^\circ + \cos^2 75^\circ}$

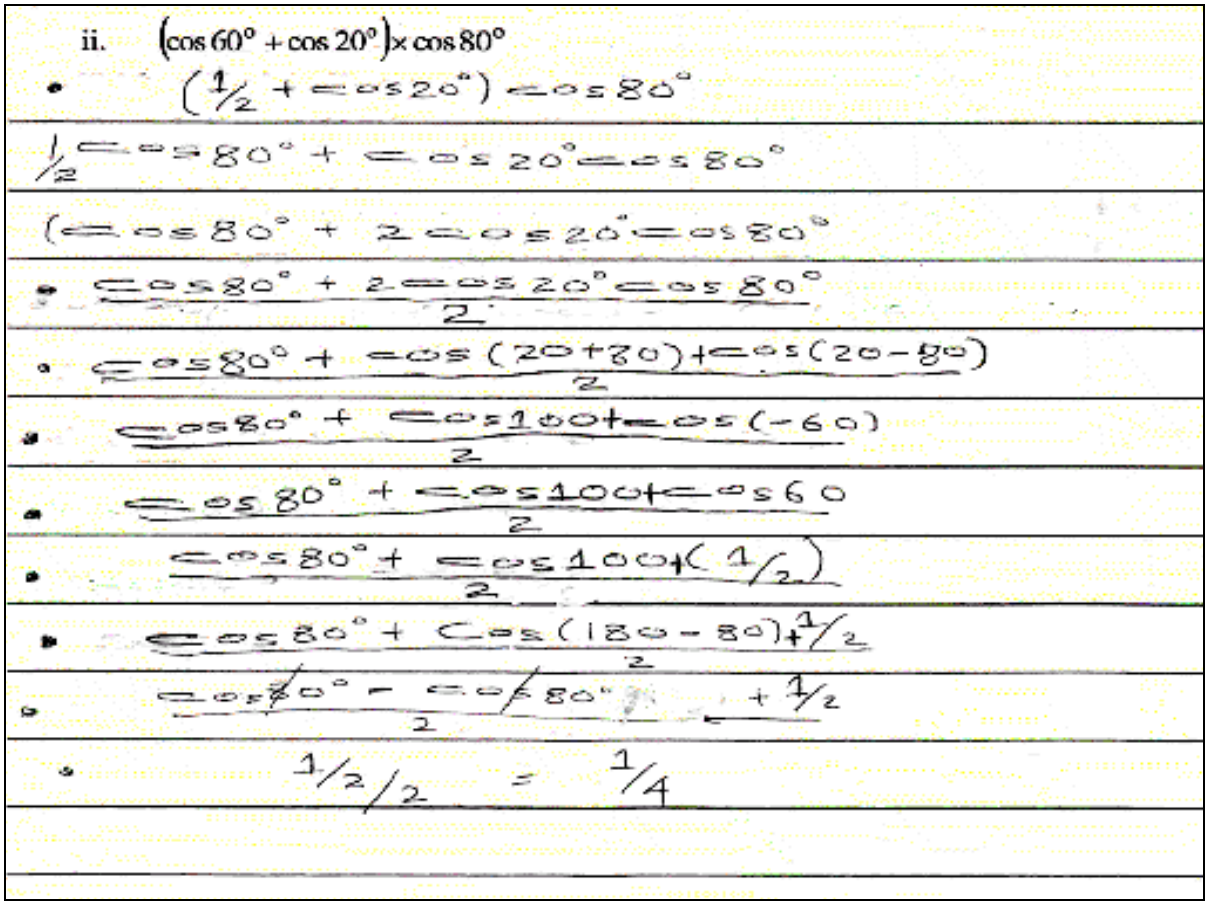
ii. $\left(\cos 60^\circ + \cos 20^\circ\right) \times \cos 80^\circ$

Better responses exhibited that candidates, in part **i**, correctly used the identity $\sin^2 75^\circ + \cos^2 75^\circ = 1$ and then applied the formula of sin*A* – sin*B* to evaluate $\dfrac{\sin 150^\circ - \sin 30^\circ}{\sin^2 75^\circ + \cos^2 75^\circ}$ in the right way.

Similarly, in part **ii**, candidates first replaced $\cos 60^\circ$ by $\dfrac{1}{2}$ and wrote $\dfrac{1}{2}\cos 80^\circ + \cos 20^\circ \cos 80^\circ$ correctly. After that, they applied the formula of $\cos A \cos B$ and simplified accordingly by applying different identities such as $\cos(180 - A)$ and $\cos(-A)$ to achieve the required answer comfortably.

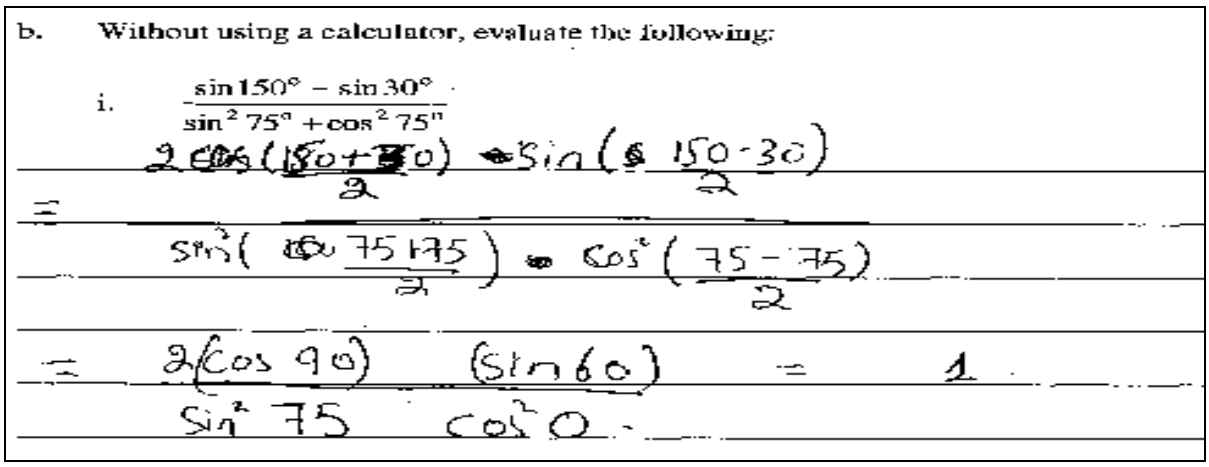
Example:

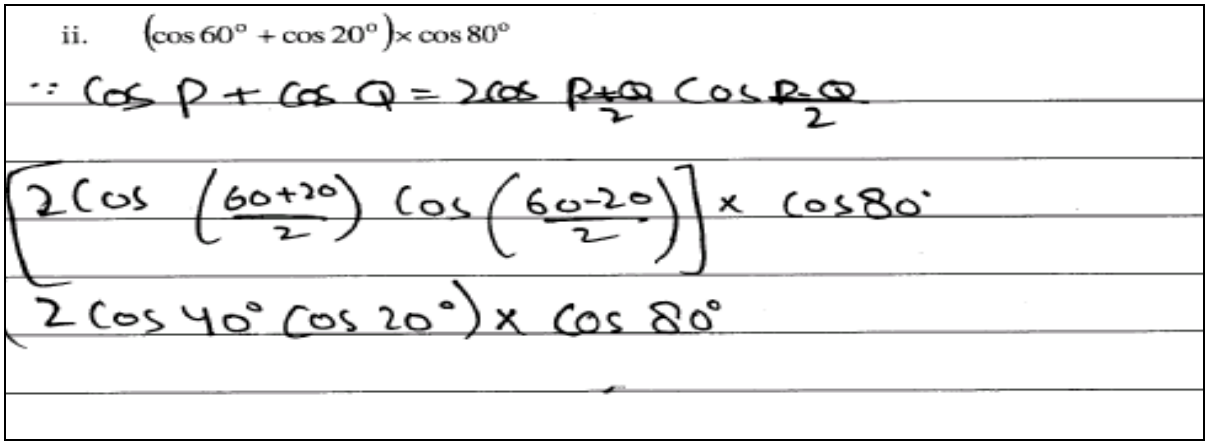




Weaker responses demonstrated that candidates, in part **i**, even though used the formula correctly but failed to evaluate because they wrote $\cos 90 = 1$ instead of $\cos 90 = 0$. It was also evident that fewer candidates were unable to write 1 in place of $\sin^2 75^\circ + \cos^2 75^\circ$ and after that they were stuck in further steps which resulted in the loss of marks. Similarly, in part **ii**, candidates did not understand the way question will be solved. They directly applied the formula of cos *A* + cos *B* to evaluate $\cos 60^\circ + \cos 20^\circ$. Because of this, they got perplexed in further steps and could not reach the required answer. It was noticed that majority of the candidates were unable to get the right steps accordingly which resulted in the loss of marks.

Example:





Question 7c:

i. Express $\dfrac{s(s-a)(s-b)(s-c)}{(a+b+c)(abc)}$ in terms of *r* and *R*.

ii. A circle is inscribed in a triangle having radius *r* units. It is given that the area of circle and perimeter of triangle are 81π square units and 48 units respectively. Use $r = \dfrac{\Delta}{s}$ to find Δ.

Better responses revealed that candidates, in part **i**, used the formula $\Delta = \sqrt{s(s-a)(s-b)(s-c)}$, $s = \dfrac{a+b+c}{2}$ and $R = \dfrac{abc}{4\Delta}$ perfectly to express $\dfrac{s(s-a)(s-b)(s-c)}{(a+b+c)(abc)}$ in terms of *r* and *R* very comfortably. This showed candidates' ability and skilful approach in getting the answer.

In part **ii**, they used the condition of area of circle and perimeter of triangle accordingly to get the correct value of *r* and *s*. This helped candidates to achieve the value of Δ as formula was given in the question as well.

Example:

i. Express $\frac{s(s-a)(s-b)(s-c)}{(a+b+c)(abc)}$ in terms of r and R. (4 Marks)

(**Note**: Symbols have their usual meanings.)

$\frac{s(s-a)(s-b)(s-c)}{(a+b+c)(abc)}$ $\quad \therefore \Delta^2 = s(s-a)(s-b)(s-c)$

$\therefore s = \frac{a+b+c}{2}$

$= \frac{(\Delta^2)}{(2s)(R+\Delta)}$ $\quad 2s = a+b+c$

$\therefore R = \frac{abc}{4\Delta}$

$= \frac{\Delta}{8sR}$ $\quad R4\Delta = abc$

$\therefore r = \frac{\Delta}{s}$

$= \frac{r}{8R}$

ii. A circle is inscribed in a triangle having radius r units. It is given that the area of a circle and perimeter of a triangle are 81π square units and 48 units respectively.

Use $r = \frac{\Delta}{s}$ to find Δ. (3 Marks)

Given $\pi r^2 = 81\pi$ —(1) $\quad a+b+c = 48$ units —(2)

Using (1) $\quad$ using (2)

$r^2 = \frac{81\pi}{\pi}$ $\quad$ as we know $(a+b+c = 2s)$

$r^2 = 81$ $\quad 2s = 48$

$r = 9$ $\quad s = \frac{48}{2} \Rightarrow s = 24$

$r = \frac{\Delta}{s}$

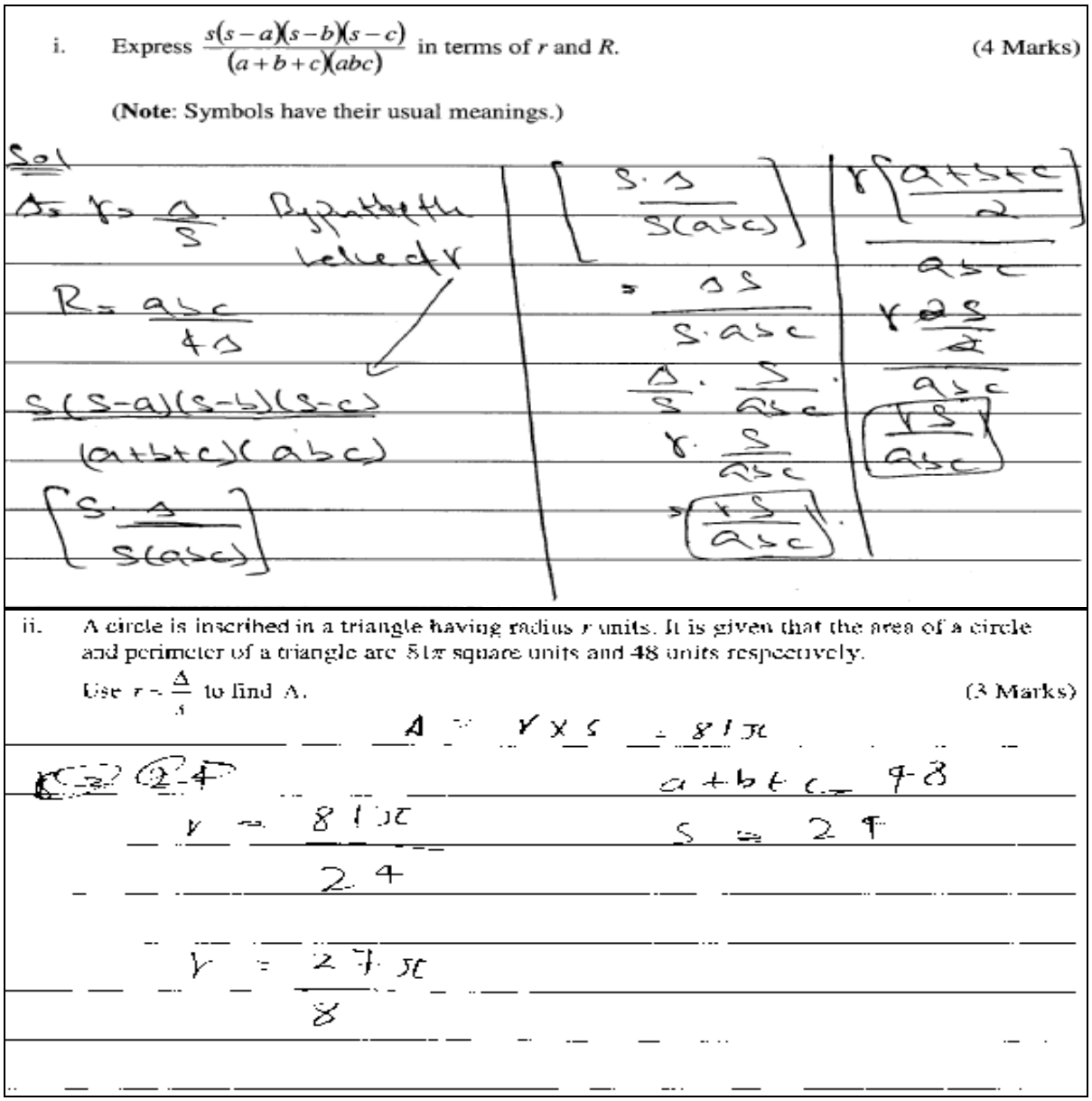
$\Delta = 216$

$9 = \frac{\Delta}{24}$

Weaker responses reflected that the candidates, in part **i**, failed to use the formula accordingly so that they could not obtain the desired answer.

In part **ii**, it is evident that candidates chose the formula of area of triangle $\Delta = \sqrt{s(s-a)(s-b)(s-c)}$ which is clear indication of lack of understanding and practice. Some candidates wrote $Area = 2\pi r$ for the area of circle which was actually incorrect and as a result, they failed to proceed further in finding the required value of Δ.

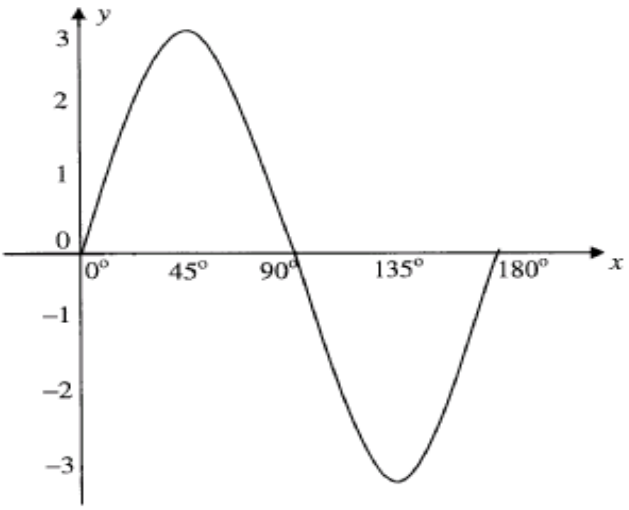
Example:



Better responses indicated that candidates read the graph very well to find the required values. They showed their ability and clear understanding about the reading of graph.

Example:

a. The figure shows the graph of $y = a\sin(px)$ for $0^\circ \le x \le 180^\circ$, where a and p are positive constants.



Use the graph to state the

i. value of a and of p. (2 Marks)

$y = a\sin(px)$

a = amplitude or range

$a = 3$

$p = \frac{2\pi}{\pi}$

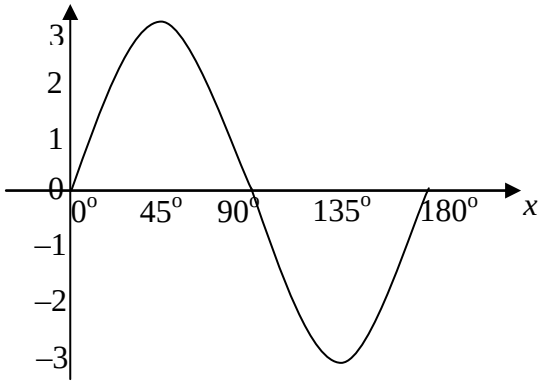
$p = 2$

ii. the period of y. (1 Mark)

Period of $y = \frac{2\pi}{2} = \pi$

Question 8:

a.



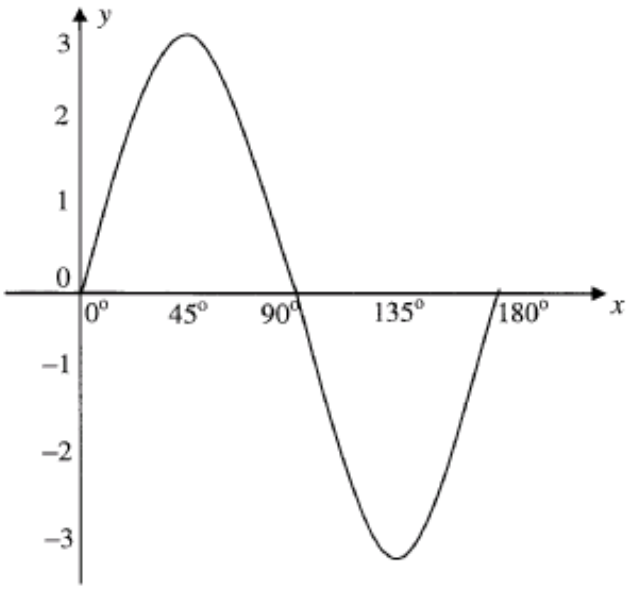
The figure shows the graph of $y = a\sin(px)$ for $0^\circ \le x \le 180^\circ$, where a and p are positive constants. Use the graph to state the

i. value of a and of p.
ii. the period of y.

Weaker responses exhibited that candidates found the incorrect value of a and of p . The candidates were clueless in reading the given graph. As a result, they did not find the period of the given function.

Example:

a. The figure shows the graph of $y = a\sin(px)$ for $0^\circ \le x \le 180^\circ$, where a and p are positive constants.



Use the graph to state the

i. value of a and of p. (2 Marks)

$a = 3$

$p = 45^\circ$

ii. the period of y. (1 Mark)

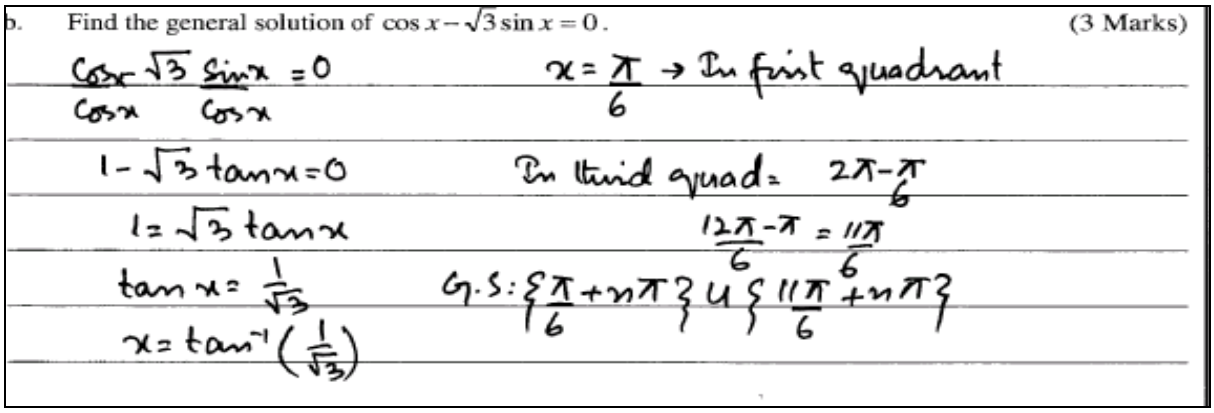
2π

Question 8b:

Find the general solution of $\cos x - \sqrt{3}\sin x = 0$.

Better responses exhibited that candidates have good understanding of concept of solution of trigonometric equations. They converted the given trigonometric equation in tan *x*. This helped them in proceeding further to get the answer. Finally, they wrote the required solution set.

Example:



Weaker responses showed that candidates failed to apply the correct technique to solve the given equation. In a few cases, candidates were able to find the principal angle; however, they failed to find the other angle satisfying the given trigonometric equation and consequently, failed to write the solution set of the equation.

Example:

